# Newton's Laws of Motion

1. $v = u - at \Rightarrow$

$v^2 = u^2 - 2as \Rightarrow 0 = (2)^2 - 2a \times 10$

$\therefore a = \frac{4}{2 \times 10} = \frac{2}{10}\ m/s^2$

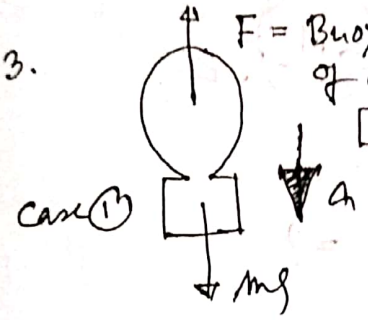
Force $= m \times a = 100 \times \frac{2}{10} = \underline{\underline{20N}}$

2. $v = u - at \Rightarrow 0 = 10 - a \times 0.1$

$a = \frac{10}{0.1} = 100\ m/s^2$

$F = ma = 0.1 \times 100 = \underline{\underline{10N}}$

3.



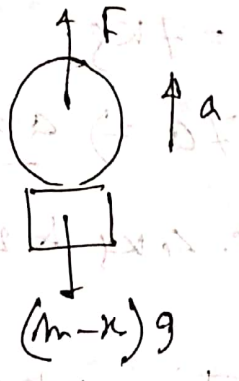

case (1)

by Newton's 2nd law

$mg - F = ma$ — (1)

Let man dropped from ballon be 'x'

case (2)




by 2nd law

$F - (m-x)g = (m-x)a$ — (2)

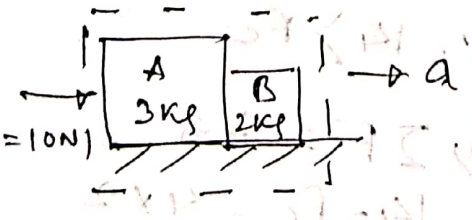
(1) + (2)

$xg = 2ma - xa$

$x(g+a) = 2ma$

$\boxed{x = \frac{2ma}{g+a}}$

(4)

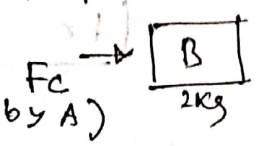



i) Consider A+B as a system,

$F = (m_A + m_B)a$

$10 = (3+2)a \Rightarrow \underline{\underline{a = 2m/s^2}}$

Consider f.b.d of 'B'




$F_c = m_B a$

a, $F_c = 2 \times 2 = 4N$

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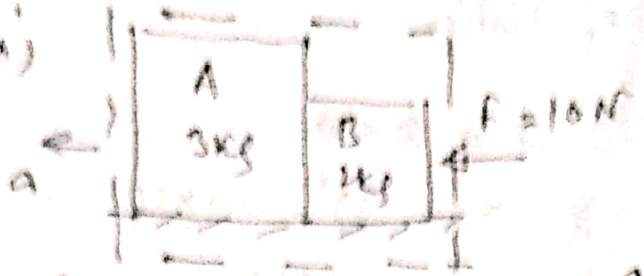
(ii)

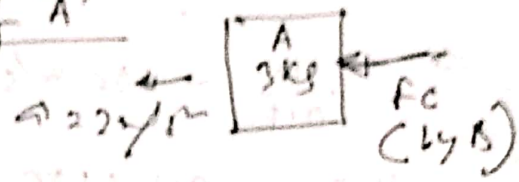



Common accl^n of A+B, $a = \frac{F}{m_A + m_B} = \frac{10}{5}$

$a = 2 m/s^2$

To find 'Fc' betn A & B

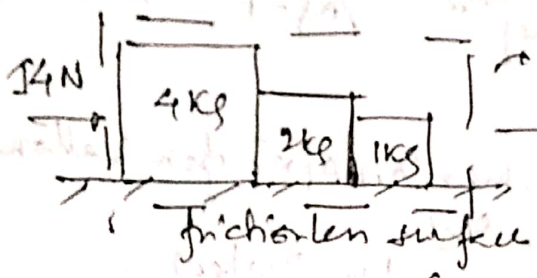
Consider f.b.d of 'A'




$\therefore F_c = m_A \times a$

$= 3 \times 2 = 6N$

5



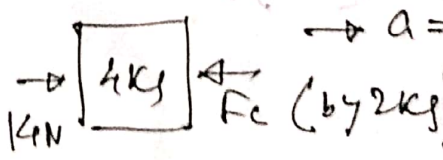

for system of mass (4+2+1) = 7kg.

$F = ma \Rightarrow 14 = 7a \Rightarrow a = 2 m/s^2$

To find contact force betn. 4kg & 2kg

Consider 4kg,

f.b.d. of 4kg




$\therefore$ 'a' to right $\therefore 14 > F_c$.

by 2nd law for 4kg, $\Sigma F = ma$

$14 - F_c = 4 \times 2$

$F_c = 14 - 8$

$F_c = 6N$

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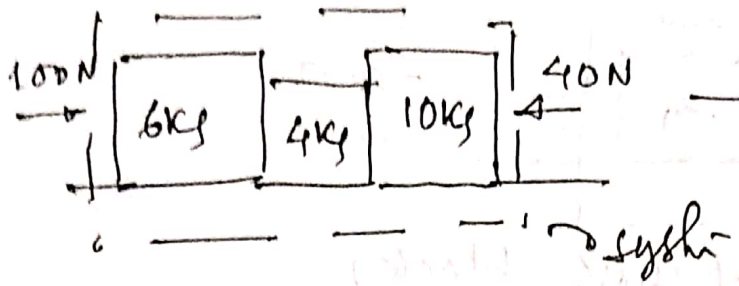
6/




→ a (common accl^n of system)

For system (6+4+10 = 20kg)

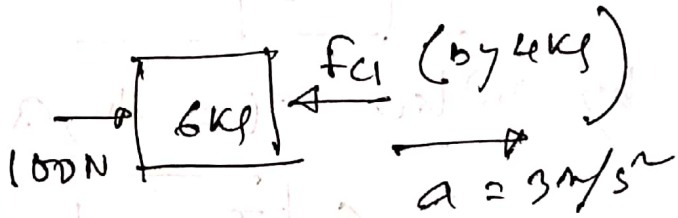
(i) $\Sigma F = ma$

$100 - 40 = 20 \times a \rightarrow a = \frac{60}{20} = 3 m/s^2$

(ii) To find $F_{c1}$ betn 6kg & 4kg

Consider 6kg,

f.b.d of 6kg



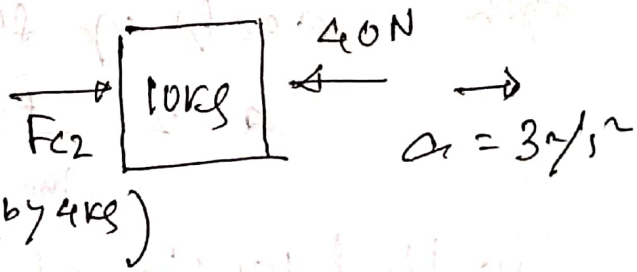

by 2nd law, $\Sigma F = ma$ → $100 - F_{c1} = 6 \times 3$

$F_c = 100 - 18 = 82 N$

(iii) To find $F_{c2}$ betn 4kg & 10kg

Consider 10kg,

f.b.d of 10kg




by 2nd law. $\Sigma F = ma$

$F_{c2} - 40 = 10 \times 3$

$F_{c2} = 70 N$

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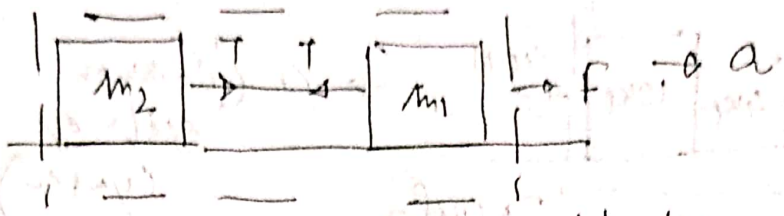
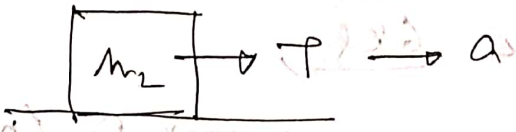
7.




Common accln of both blocks

$$a = \frac{F}{m_1+m_2} \quad (1)$$

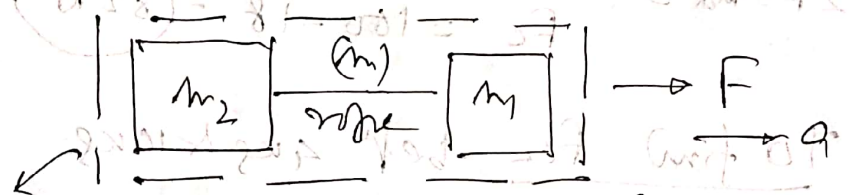
To find tension T in rope
(∵ massless, so tension is same at every pt) consider f.b.d of $m_2$,




by 2nd law $T = m_2 a$ (1)

$$a \quad \boxed{T = \frac{m_2 F}{m_1+m}}$$

8.

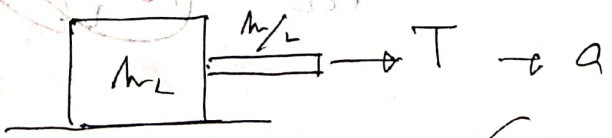



syst

accln of system $(m_1+m_2+m)$

$$a = \frac{F}{m_1+m_2+m} \quad (1)$$

To find tension at mid pt of rope
Consider f.b.d of $m_2 + \frac{m}{2}$




by 2nd law $T = \left(m_2 + \frac{m}{2}\right) a$ (2)

$$\therefore \boxed{T = \left(m_2 + \frac{m}{2}\right) \frac{F}{m_1+m_2+m}}$$

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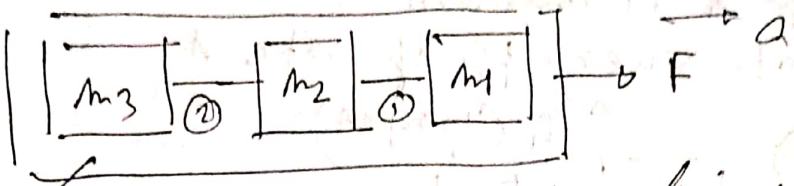
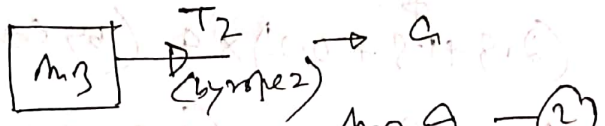
9.




common accl of system $(m_1+m_2+m_3)$

is $a = \dfrac{F}{m_1+m_2+m_3}$ — (1)

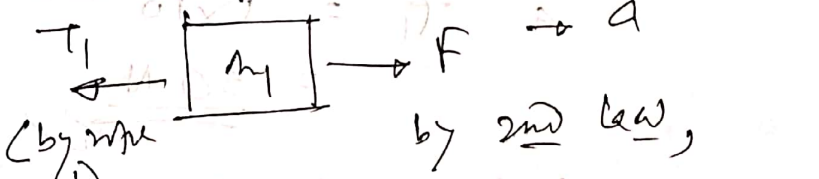
for $T_2$, consider f.b.d of $m_3$




by 2nd law, $T_2 = m_3 a$ — (2)

$$\boxed{T_2 = \frac{m_3 F}{m_1+m_2+m_3}}$$

for $T_1$, consider f.b.d of $m_1$




by 2nd law,

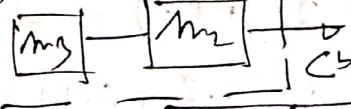
$F - T_1 = m_1 a$ — (3)

$$F - T_1 = \frac{m_1 F}{m_1+m_2+m_3}$$

$$T_1 = F - \frac{m_1}{m_1+m_2+m_3}$$

$$\boxed{T_1 = \frac{(m_2+m_3)F}{m_1+m_2+m_3}}$$

Alternative: to find $T_1$, consider f.b.d of $m_2+m_3$ as system




by 2nd law, $T_1 = (m_2+m_3)a$, $\boxed{T_1 = \dfrac{(m_2+m_3)F}{m_1+m_2+m_3}}$

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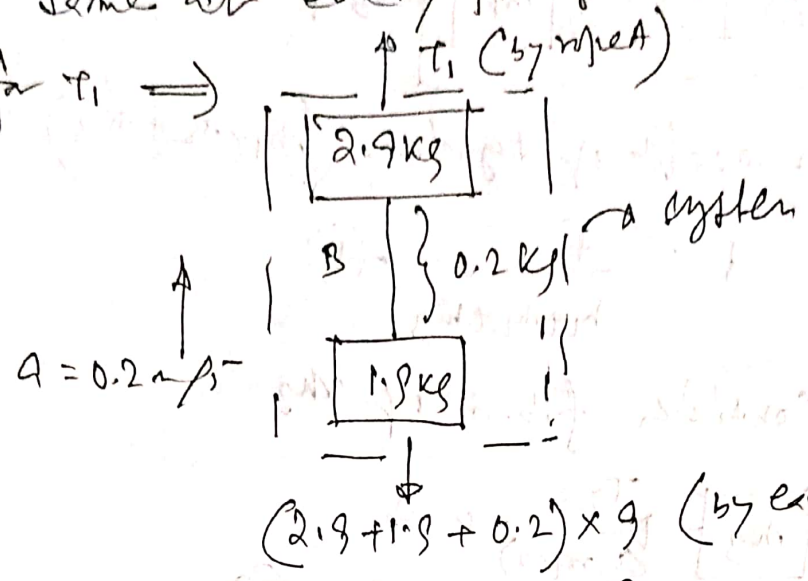
10/ ∵ rope A is massless so tension is same at every pt of rope A,

for $T_1$ ⟹




by 2nd law for motion of (2.9 + 0.2 + 1.9) kg

$$T_1 - (2.9+0.2+1.9)g = (2.9+0.2+1.9)a$$

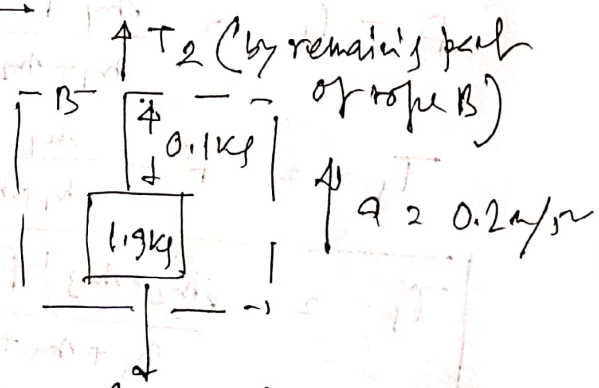
$$T_1 = 5(g+a) = 5(9.8+0.2)$$

$$\therefore T_1 = 5\times 10$$

$$= \underline{50\ N}$$

for tension at mid pt of rope B.

Consider f.b.d of half of rope B & (1.9 kg) as a single system →




by 2nd law;

$$\cancel{T_2 - 20 = (1.9+0.1)a}$$

$$\cancel{T_2 = 20 + 2a}$$

$$T_2 - 2g = (1.9+0.1)a$$

$$T_2 = 2(g+a)$$

$$= 2(9.8+0.2)$$

$$= \underline{20N}$$

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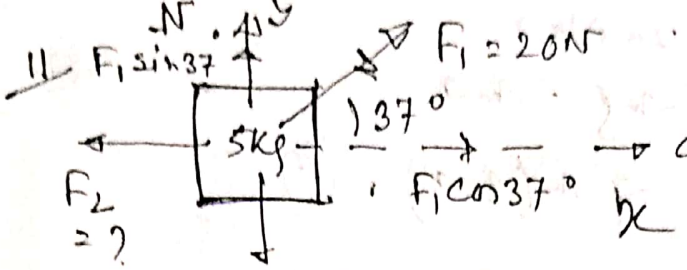
11. f.b.d of 5kg



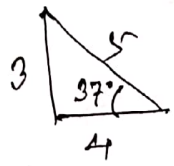

along y axis there is no accl. ∴ $\Sigma F_y = 0$

i.e $N + F_1 \sin 37°$ balances wt 5g.

along x axis, there is accl $a = 2 m/s^2$

∴ by eqn, $\Sigma F_x = m a_x$

$$(F_1 \cos 37° - F_2) = 5 \times 2$$




$$\cancel{20}^{4} \times \frac{4}{\cancel{5}} - F_2 = 10$$

$$F_2 = 16 - 10$$

$$\boxed{F_2 = 6N}$$

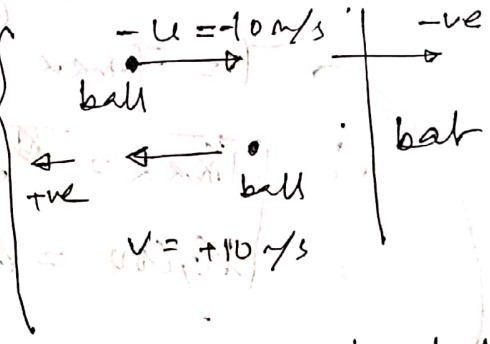
12. $F = ma$

$$F = m\left(\frac{v+u}{t}\right)$$

$$= 0.25 \times \frac{(10+10)}{0.01}$$

$$= 25 \times 20$$

$$= 500 N$$




$$a = \frac{\text{change in velocity}}{\text{time}}$$

$$= \frac{v - (-u)}{t}$$

$$= \frac{v+u}{t}$$

Note veloc is vector, in one-dimensional motion if we choose velocity towards right (+ve), then velocity towards left is -ve.

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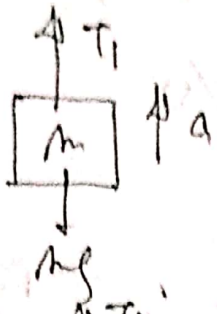
13.

C-1

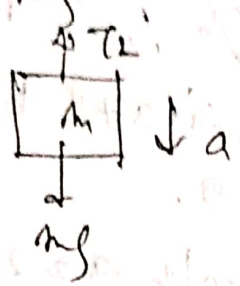



2nd law

$T_1 - mg = ma$

$T_1 = m(g+a)$ — (1)

C-2




2nd law

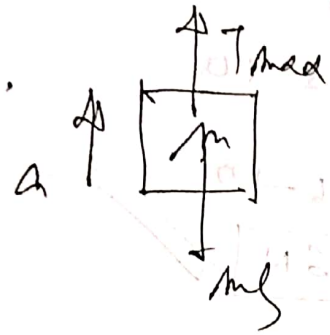
$mg - T_2 = ma$

$T_2 = m(g-a)$ — (2)

$\therefore$ (1) ÷ (2) $\frac{T_1}{T_2} = \frac{g+a}{g-a}$

$\frac{T_1}{T_2} = \frac{10+5}{10-5} = \frac{3}{1}$

14.




2nd law

$T - mg = ma$

$T = m(g+a)$

$\because$ g, m are const $\therefore$

$T = T_{max}$ when $a = a_{max}$

$\therefore$ $\cancel{T_{max} = m(g + a_{max})}$

$\cancel{10{,}000 = 800(10 + a_{max})}$

back to

$T_{max} = m(g + a_{max})$

$10000 = 800(10 + a_{max})$

$a_{max} = \frac{100-80}{8} = \frac{20}{8} = \frac{5}{2}$

$= 2.5\ m/s^2$

(we have considered $g = 10\ m/s^2$)

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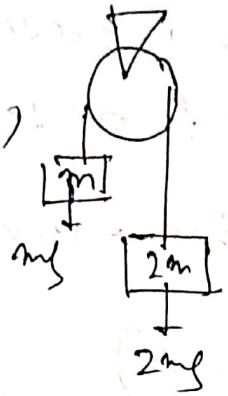
15.

(i)



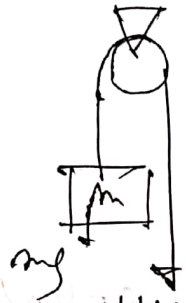

accl of system i.e accl of m

$= \frac{\text{net pulling force}}{\text{total mass}}$

$= \frac{2mg - mg}{2m + m} = \frac{mg}{3m}$

$= \frac{g}{3}$ ↓ (downward)

(ii)




applied force = 2mg

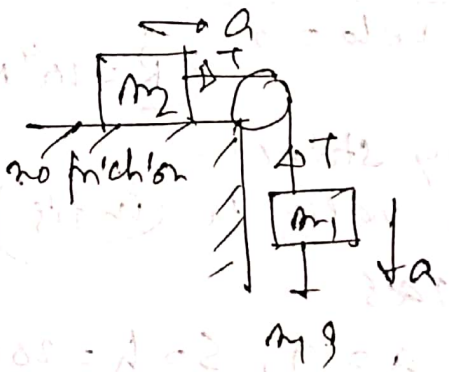
accl of m

$= \frac{\text{net pulling force}}{\text{total mass}}$

$= \frac{2mg - mg}{m} = \frac{mg}{m}$

$= g$ ↑ (upward)

16/

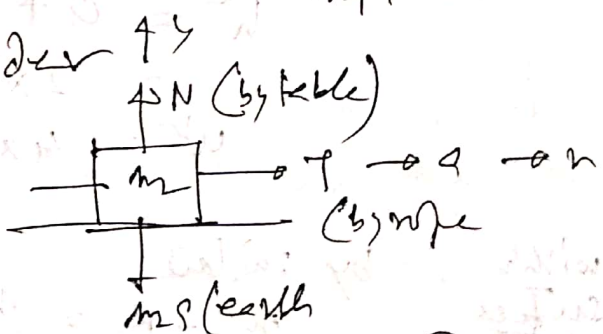



accl of $m_1$ & $m_2$

$a = \frac{\text{net pulling force}}{\text{total mass}}$

$a = \frac{m_1 g}{m_1 + m_2}$ — (1)

To find T consider f.b.d of $m_2$ →




∵ No accl along y,

∴ N balances $m_2 g$.

But along x by 2nd law $T = m_2 a$ — (2)

$T = m_2 \times \frac{m_1 g}{m_1 + m_2}$

$= m_1 m_2 g / (m_1 + m_2)$

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17. Force = change in momentum in one sec

F = change in (mass × velo) in one sec

= mass in one sec × change in velo

alt

$F = v\frac{dm}{dt}$

$\frac{dm}{dt} = 0.25$,

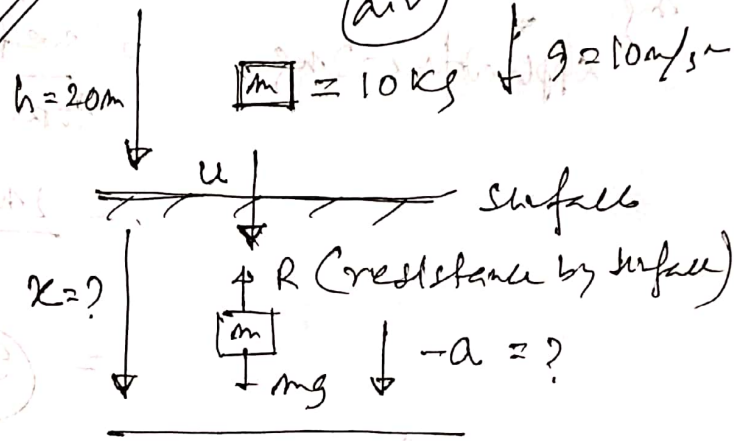
$v = 1000\ m/s$

$\therefore F = 250\ N$

$= 0.25 \times (1000 - 0)$

$= 250\ N$

18. velo = 0 (air)




($R = 10^4\ N$)

<u>in air</u> let the body strikes the surface with velo. u (in ft/s)

apply $v^2 = u^2 + 2as$

here $v = u$, $u = 0$, $a = g$, $s = h = 20\ m$

$u^2 = 0 + 2 \times gh$

$= 2 \times 10 \times 20$

$u^2 = 4 \times 100 \quad \therefore u = 20\ m/s$

<u>within surface</u> by 2nd law

$mg - R = m(-a)$

$\therefore R = m(g + a)$ — (1)

within surface the block has -ve accl.

18 cont'd

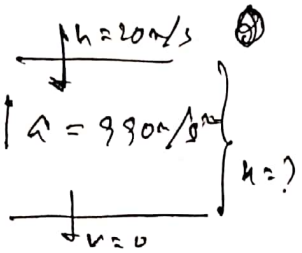
$$10000 = 10(10 + a)$$

$$a = \frac{9900}{10} = \underline{990\ m/s^2} \text{ (retardation)}$$

So within surface, apply

$$v^2 = u^2 - 2as$$

$v = 0$, $u = 20\ m/s$, $a = 990\ m/s^2$, $s = x$ ?

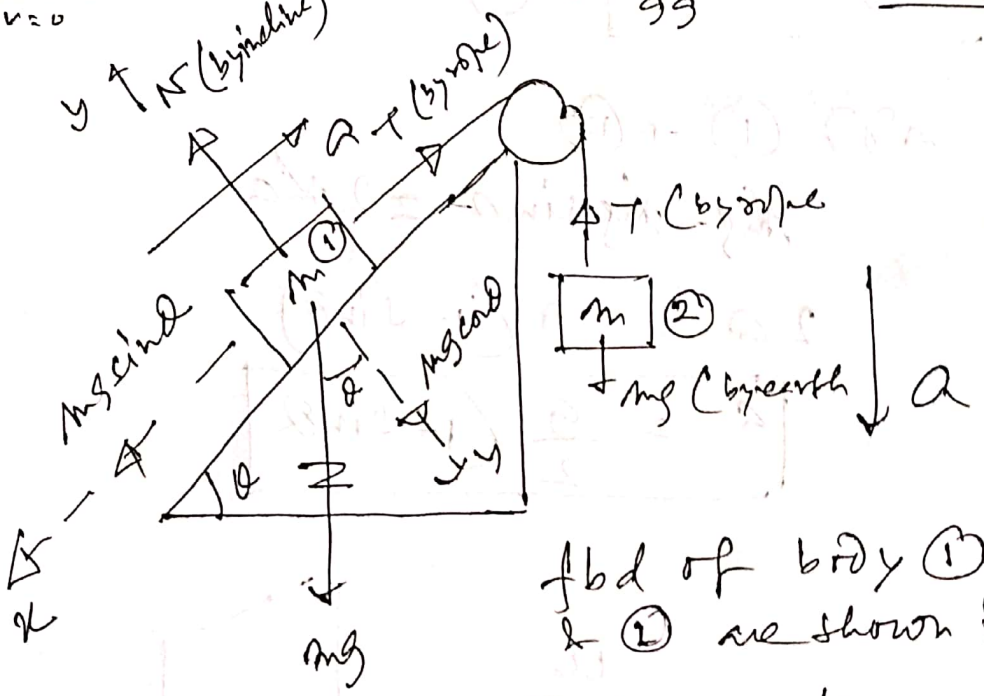



$$0 = u^2 - 2ax$$

$$x = \frac{u^2}{2a} = \frac{20 \times \cancel{20}}{\cancel{2} \times 99\cancel{0}}$$

$$= \frac{20}{99} = \underline{\underline{0.2\ m}}$$

19




fbd of body ① & ② are shown in fig

[for body 1 there is no accl along y axis so N balances mg cosθ, & so N, mg cosθ are useless in this problem]

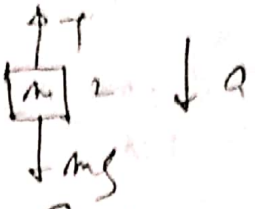
for body 2, pulling force is mg, & tension in rope is T upward, for body 1, pulling force down the incline is mg sinθ & tension in rope is T up along incline,

Since $\sin\theta < 1$ $[\because \theta < 90°]$

$\therefore$ $mg \sin\theta < mg$, hence body ② moves down with accl a & body ① moves

upward. with same accl^n 'a'

So for body (2)

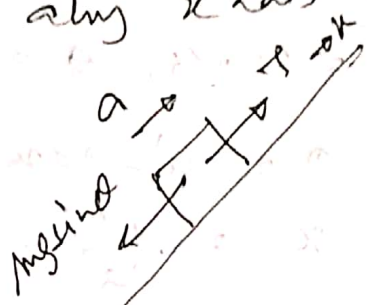



by 2nd law.

$$\boxed{mg - T = ma} \quad \text{—(1)}$$

& for body (1) along x axis, i.e along incline,

by 2nd law




$$\boxed{T - mg\sin\theta = ma} \quad \text{—(2)}$$

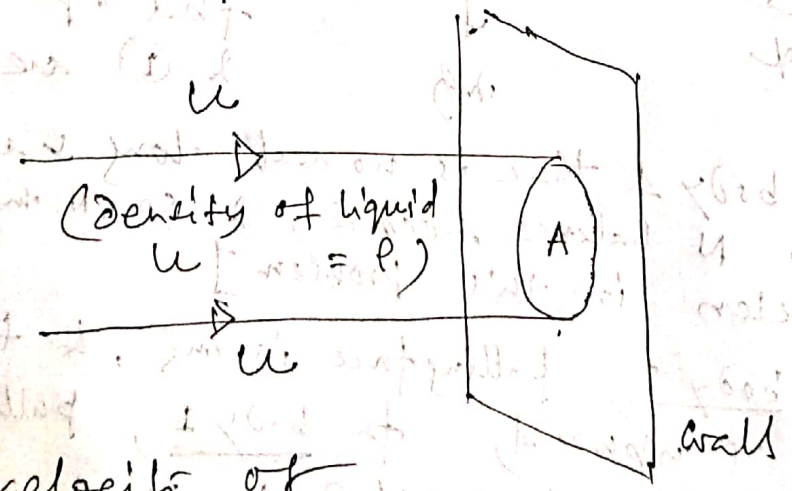
add (1) + (2)

$$\cancel{mg} - \cancel{mg}\sin\theta = 2\cancel{m}a$$

$$2a = g(1 - \sin\theta)$$

$$\boxed{a = \frac{g}{2}(1 - \sin\theta)}$$

20.




Final velocity of water jet after hitting wall = 0

initial velocity of water jet = u

So change in velo of water jet = u

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Force by water jet on wall

= Force on waterjet by wall (by 3rd law)

= Change in (momentum) of water in 1 sec (by 2nd law)

= Change in (mass × velo) of water in 1 sec

$F$ = mass of water in 1 sec × change in its velocity

[ mass of water in 1 sec

$= \frac{M}{t} = \frac{\text{Area} \times \text{length} \times \text{density}}{\text{time}}$

$= \text{Area} \times \text{velocity} \times \text{density}$

$= A u \rho$ ]

$\therefore F = A u \rho \times u$

$$\boxed{F = A u^2 \rho}$$

$= 10^{-2} \times (15)^2 \times 1000$

$= 225 \times 10 = \underline{2250\ N}$

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